Geometrical Patchwork: A Gluing Construction of Einstein Metrics via High-Dimensional Dehn Filling

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Contents

Abstract

This dissertation studies an analytic construction of negative Einstein metrics through a *cutting and gluing* procedure (referred as Dehn Filling). We begin by analysing the construction of [And06] and identifying its limitations, which will motivate a discussion of an alternative proof by [Bam12]. The final part of the dissertation explores recent novel numerical techniques for approximating these metrics using neural networks.

Acknowledgments

I would like to sincerely thank my supervisor for his guidance and support in preparation of this dissertation.

Chapter 1

Introduction

This dissertation aims to study an analytic construction of negative Einstein metrics using a cutting and gluing procedure. Broadly speaking, we begin with a finite-volume hyperbolic manifold $(M_{\mathrm{hyp}}, g_{\mathrm{hyp}})$ with a finite number of cusps, each bounded by a torus T^{n-1}. The process involves *cutting* the manifold along these toroidal boundaries and isometrically *gluing* solid tori equipped with a well-chosen Einstein metric (in our setting, the black-hole metric) to a simple closed geodesic $\sigma \in T^{n-1}$. This process is known as *Dehn Filling.* This results in an approximate Einstein metric g_σ on the resulting manifold M_σ, which is Einstein everywhere except in the gluing regions. Then, one aims to apply the Implicit Function Theorem to the so-called *Einstein operator* to *perturb* this approximate solution into an exact one. To achieve this, we require a uniform bound on the inverse of the linearized Einstein operator, denoted as L. This is the main technical challenge of the proof.

More formally, the statement we aim to prove is:

Theorem 1. *[And06; Bam12] Let (M_{hyp}, g_{hyp}) be a complete, non-compact hyperbolic n-manifold of finite volume, $n \geq 3$, with toral ends. Then any manifold M_σ obtained by a sufficiently large Dehn filling of the ends of M_{hyp} admits a complete, finite volume Einstein metric g, of uniformly bounded curvature.*

Moreover, the metrics on M_σ can be constructed in such a way so that he sequence (M_σ, g_σ) converges to the initial hyperbolic manifold (M_{hyp}, g_{hyp}) in the pointed Gromov-Hausdorff sense if the basepoints are chosen away from the cusps.

This theorem was proven by Anderson [And06] in 2003, and in 2011 Bamler [Bam12] provided an alternative proof. [And06] could only prove a uniform bound on L^{-1} whenever the cutting and gluing operations at different cusps are sufficiently controlled relative to one another. [Bam12] gets around this difficulty by noticing that the bad invertibility of L arises from a particular class of deformation of the cusp metric, referred to as *Einstein trivial variations.* By choosing a norm that treats these variations differently, we can get a uniform bound on L^{-1}, independent of the gluing construction.

We just presented the main intuitions behind the proof, the details will be formalized in this essay. In particular, the structure of this dissertation is as follows:

In Chapter 2, we begin by introducing the relevant background theory on hyperbolic manifolds. Chapter 3 describes the construction of the approximate Einstein metric via the gluing procedure. In Chapter 4, we discuss how to perturb this approximate metric into an exact one. We first analyse the limitations of Anderson's approach [And06] in Subsection 4.2, which motivates the discussion of Bahmler's alternative proof [Bam12] in Subsection 4.3.

Chapter 5 focuses on recent numerical methods to approximate these metrics using physics-informed neural networks. In particular, we explore the structure of the loss surface in such architectures and its connection to the stability of Einstein metrics.

Chapter 6 provides concluding remarks on the topic.

Chapter 2

Background theory

2.1 Hyperbolic geometry

Since Einstein manifolds are the primary objects of study in this essay, we begin by introducing their definition.

Definition 1 (Einstein Manifold [L.B87]). *An Einstein manifold is a Riemannian manifold (M, g) such that the Ricci tensor is proportional to the metric tensor:*

$$\mathrm{Ric} = \lambda g \tag{2.1}$$

for some constant $\lambda \in \mathbb{R}$. Such a metric g is called an Einstein metric.

This definition is well-posed because both the Ricci curvature and the Riemannian metric are symmetric (0,2)-tensors, so we can compare them. As we can always normalize, we are interested in the sign of the constant λ. In this essay, we will be looking at the case when λ is negative. Einstein metrics have historical importance in Mathematical Physics as they arise as solutions to the vacuum Einstein field equations [L.B87].

In this essay, we will be working with hyperbolic manifolds for all of our constructions. A hyperbolic manifold is a complete Riemannian manifold locally isometric to the hyperbolic space $\mathbb{H}^n$ (that is, with sectional curvature -1). One can prove, see for example [Mar16], that every complete hyperbolic n-manifold with boundary is the quotient of $\mathbb{H}^n$ by some subgroup $\Gamma < \mathrm{Isom}(\mathbb{H}^n)$ acting freely and properly discontinuously. Remember that the action of $\Gamma < \mathrm{Isom}(\mathbb{H}^n)$ is free if the identity element on Γ is the only element to have a fixed point in $\mathbb{H}^n$, and is properly discontinuous if it contains no sequence of distinct elements converging to the identity element. These properties ensure that the resulting quotient manifold is smooth, the quotient is well-defined and there are no orbifold-like singularities. $\mathrm{Isom}(\mathbb{H}^n)$ can be classified into three types, depending on the number of fixed points:

- **Elliptic**: Has at least one fixed point in $\mathbb{H}^n$. Can be thought of as a rotation around a point in hyperbolic space.
- **Parabolic**: Has no fixed points in $\mathbb{H}^n$ but fixes a single point in $\partial\mathbb{H}^n$. This fixed point is the center of a horosphere.
- **Hyperbolic**: Has no fixed points in $\mathbb{H}^n$, but fixes two points on $\partial\mathbb{H}^n$. It acts as a translation along the geodesic connecting these two points.

Hence, we get that a group $\Gamma < \text{Isom}(\mathbb{H}^n)$ acts freely if and only if it does not contain elliptic isometries.

Remark 2.1.1. In dimension 3, $\Gamma = \text{PSL}(2, \mathbb{C}^2)$ and the action on an element of the boundary $z \in \partial\mathbb{H}^3$ is given by a Möbius Transformation.

2.2 Thick-thin decomposition of hyperbolic manifolds

A key step in the cutting and gluing construction described Chapter 3 will be the ability to decompose the manifold into a *thick* and a *thin* part. Before introducing this decomposition, we first need to present some definitions.

Definition 2 (Cusps). *Let M be a compact flat $(n-1)-$manifold. A cusp N is a hyperbolic manifold diffeomorphic to $\mathbb{R}_+ \times M$ with wraped product metric*

$$g = dt^2 + e^{-2t}g_0,$$

where g_0 is the flat metric on M.

By Bieberbach theorem ([Cha86]), any compact, flat Riemannian manifold M is finitely covered by a torus. Therefore, $M = T^{n-1}/\Gamma$ for finite groups Γ of Euclidean isometries acting freely in T^{n-1}. In this dissertation, we will only be working with the case where $\Gamma = \{1\}$, as this can always be achieved by passing to a finite covering space of N with covering group Γ [CF94]. Although cusps are regions in the manifold that extend to infinity, they have finite volume due to the exponential decay of the metric, as proved in Lemma 2.2.1.

Lemma 2.2.1. *Cusps from Definition 2 have finite volume.*

Proof. Let N be a cusp of the form $[0, \infty) \times T^{n-1}$ with the warped product metric

$$g = dt^2 + e^{-2t}g_0,$$

where g_0 is a flat metric on the $(n-1)$ dimensional torus T^{n-1} and t ranges from $[0,\infty)$. We then have $\sqrt{\det(g)} = e^{-t(n-1)}$, so the volume of the cusp N is given by:

$$\mathrm{Vol}(N) = \int_0^\infty \left(\int_{T^{n-1}} e^{-t(n-1)}\, dV_0 \right) dt,$$

where dV_0 is the volume element of T^{n-1}. We then get

$$\mathrm{Vol}(N) = \mathrm{Vol}(T^{n-1}) \int_0^\infty e^{-t(n-1)}\, dt$$

and

$$\mathrm{Vol}(N) = \frac{1}{n-1} \cdot \mathrm{Vol}(T^{n-1}).$$

□

Definition 3 (R-Tube [Mar16]). *An R-tube, denoted $\mathcal{T}$, is the quotient $N_R(I)/\Gamma$, where $N_R(I)$ is the closed R-neighbourhood around a geodesic I, and $\Gamma \cong \mathbb{Z}$ (i.e., the cylindrical neighbourhoods around geodesics). These tubes are bounded, and if the manifold M is orientable, then $\mathcal{T} \cong D^{n-1} \times S^1$.*

Definition 4 (Injectivity radius [Pur20]). *Let $M = \mathbb{H}^n/\Gamma$ be a complete hyperbolic manifold and $\pi : \mathbb{H}^n \to M$ the projection from the universal cover. For every $x \in M$, we have*

$$\mathrm{injrad}(x) = \frac{1}{2} d(\pi^{-1}(x)),$$

where $d(\pi^{-1}(x))$ is the infimum of $d(x,y)$ among all pairs x, y of distinct points in $\pi^{-1}(x)$. Alternatively, $\mathrm{injrad}(x)$ is the supremal radius $r > 0$ such that a metric r-ball centered x is embedded.

Finally, we are ready to introduce the thick-thin decomposition of hyperbolic manifolds:

Definition 5 (Thick-thin decomposition). *Let M be a complete hyperbolic n-manifold of finite volume, and let $\epsilon > 0$. Define the ϵ-thin part of M, denoted $M^{<\epsilon}$, to be*

$$M^{<\epsilon} = \{x \in M \mid \mathrm{injrad}(x) < \epsilon/2\}.$$

Similarly, the ϵ-thick part, denoted $M^{\geq\epsilon}$, is defined to be

$$M^{\geq\epsilon} = \{x \in M \mid \mathrm{injrad}(x) \geq \epsilon/2\}.$$

In particular, $M = M^{<\epsilon} \cup M^{\geq\epsilon}$. Around 1970, Grigory Margulis proved that the structure of the components of the thin parts is actually very simple.

Theorem 2 (Margulis Lemma)**.** *There exists a universal constant $\epsilon > 0$, depending only on the dimension n, such that the ϵ-thin part of any complete hyperbolic manifold M of finite volume consists of $\epsilon-$tubes around closed geodesics and cusps. The supremum of all such constants ϵ is called Margulis constant.*

Remark 2.2.1. Margulis' result in the original paper was much more general, and it states that any discrete subgroup of a Lie group G generated by elements in a neighbourhood U of $e \in G$ is nilpotent [Mar16]. Theorem 2 corresponds to the case where $G = \mathrm{Isom}(\mathbb{H}^n)$.

The proof of this Theorem can be found in [Mar16]. A cartoon illustrating the results of Theorem 2 can be seen in Figure 2.1.

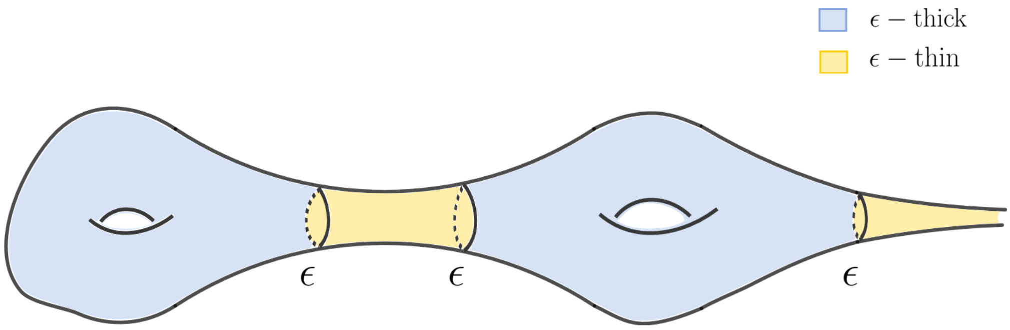


Figure 2.1: Thick-thin decomposition of a hyperbolic 3-manifold. The thin part are tubular neighbourhoods along geodesics and cusps.

Chapter 3

Construction of approximate Einstein metric

3.1 Manifold decomposition

In this chapter, we describe the construction of the approximate metric by [And06; Bam12]. Let M_{hyp} be an n-dimensional complete hyperbolic manifold with a finite number of cusps. From Theorem 2 we know we can decompose M_{hyp} into a thick part M_{thick} and a thin part M_{thin} such that

$$M_{\text{thick}} \cup \mathcal{T}_1 \cup \cdots \cup \mathcal{T}_{p'} = M_{\text{hyp}} \setminus \bigcup_{k=1}^{p} N_k,$$

where N_k are cusps and $\mathcal{T}_k$ tubes, with indices satisfying $1 \leq p' < \infty$ and $1 \leq p < \infty$. From this point onward, we will consider the core part of the manifold

$$M_{\text{core}} = M_{\text{thick}} \cup \mathcal{T}_1 \cup \cdots \cup \mathcal{T}_{p'}$$

separately from the cusps, as illustrated in Figure 3.1.

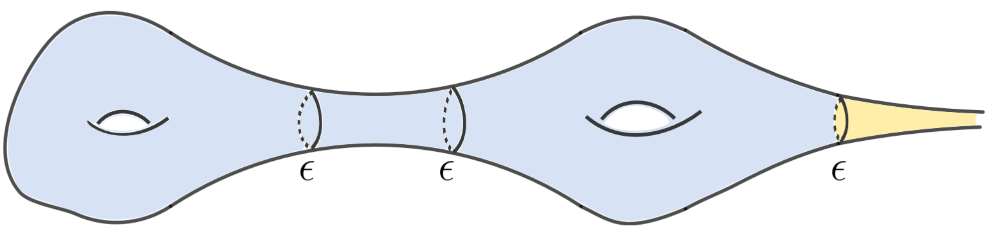


Figure 3.1: Core of the manifold consisting of the thick part and tubes (blue region) and cusp (yellow region)

We are interested in constructing a compact manifold M_σ, which is obtained by gluing together $M_{\text{core}} = M_{\text{hyp}} \setminus \bigcup_{k=1}^{p} N_k$ with $\hat{N}_k$, where $\hat{N}_k$ is diffeomorphic to a solid torus

$D^2 \times T^{n-2}$. Sections 3.1.1, 3.1.2 and 3.1.3 will describe this gluing process (called Dehn Filling) and interpolation of the metrics.

3.1.1 Metric on the core of the manifold $M_{\mathbf{core}}$

On the core of the manifold, $M_{\text{hyp}} \setminus \bigcup_{k=1}^{p} N_k$, we define the hyperbolic metric g_{hyp} as

$$g_{\text{hyp}} = r^{-2}\, dr^2 + r^2 \left(dx_2^2 + \cdots + dx_n^2\right), \tag{3.1}$$

where $(r, x^2, \ldots, x^n)$ on $\mathbb{R}^+ \times \mathbb{R}^{n-1}$. Note that as introduced in definition 2, every hyperbolic cusp is the quotient of $\mathbb{R}^+ \times \mathbb{R}^{n-1}$ with the given metric under a discrete subgroup Euclidean isometries on the last factor. Setting $r = e^{-t}$ we recover the cusp metric introduced in definition 2, $r \to 0$ giving the contracting end of the cusp.

Lemma 3.1.1. *g_{hyp} as defined in formula 3.1 is Einstein.*

Proof. To prove g_{hyp} is Einstein, we need to prove it satisfies equation 2.1. To do that, we first calculate the Christoffel symbols with

$$\Gamma^k_{ij} = \frac{1}{2} \sum_{l=1}^{n} g^{kl} \left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right). \tag{3.2}$$

In this case, the metric components are

$$g_{rr} = r^{-2}, \quad g_{ii} = r^2 \,(\text{for } i = 2, \ldots, n), \quad g_{ij} = 0 \,(\text{for } i \neq j).$$

Plugging these values into equation 3.2 we get that the nonzero Christoffel symbols are

$$\Gamma^r_{rr} = \frac{1}{2} r^2 \partial_r \frac{1}{r^2} = -\frac{1}{r}$$

$$\Gamma^r_{ii} = -\frac{1}{2} r^2 \partial_r g_{ii} = -r^3$$

$$\Gamma^i_{ri} = \Gamma^i_{ir} = \frac{1}{2r^2} \partial_r r^2 = \frac{1}{r} \;(\text{for } i = 2, \ldots, n).$$

Rather than computing the Ricci tensor explicitly via the Levi-Civita connection, we use the following equation (see [L.B87])

$$\mathrm{Ric}_{\mu\nu} = \sum_{\lambda=1}^{n} \partial_\lambda \Gamma^\lambda_{\mu\nu} - \sum_{\lambda=1}^{n} \partial_\nu \Gamma^\lambda_{\mu\lambda} + \sum_{\lambda=1}^{n} \sum_{\sigma=1}^{n} \left(\Gamma^\lambda_{\lambda\sigma} \Gamma^\sigma_{\mu\nu} - \Gamma^\lambda_{\mu\sigma} \Gamma^\sigma_{\lambda\nu}\right). \tag{3.3}$$

The only non-zero contributions to Ric_{rr} come from the terms with $\lambda = i$ and $\sigma = i$ for $i = (2, ..., n)$ getting

$$\mathrm{Ric}_{rr} = \partial_i \Gamma^i_{rr} - \partial_r \Gamma^i_{ri} + \Gamma^i_{ii}\Gamma^i_{rr} - \Gamma^i_{ri}\Gamma^i_{ir} = \mathrm{Ric}_{rr} = -(n-1) \cdot \frac{1}{r^2}.$$

A similar calculation yields that the only non-zero contributions of Ric_{ii} involve $\lambda = r$ and $\sigma = r$, getting

$$\mathrm{Ric}_{ii} = -(n-1)r^2 \quad \text{and} \quad \mathrm{Ric}_{ri} = \mathrm{Ric}_{ir} = 0.$$

To verify the Einstein condition, we check:

$$\mathrm{Ric}_{rr} = -(n-1) \cdot \frac{1}{r^2}, \quad g_{rr} = \frac{1}{r^2} \quad \Rightarrow \quad \mathrm{Ric}_{rr} = -(n-1)g_{rr},$$

$$\mathrm{Ric}_{ii} = -r^2, \quad g_{ii} = r^2 \quad \Rightarrow \quad \mathrm{Ric}_{ii} = -(n-1)g_{ii}.$$

For off-diagonal components $\mathrm{Ric}_{ij} = 0$ for $i \neq j$. Thus, the Ricci tensor satisfies $\mathrm{Ric}_{ij} = -(n-1)g_{ij}$ and the metric is Einstein.

□

3.1.2 Metric on the solid torus $\hat{N}_k$

On the solid torus $\hat{N}_k$, we equip the black-hole metric. Consider the coordinates $(r, \theta, x_3, \ldots, x_n)$ on $\mathbb{R}^n = \mathbb{R}^2 \times \mathbb{R}^{n-2}$, where (r, θ) are polar coordinates on the first factor. These coordinates range from $r \in [r', \infty)$ and $\theta \in [0, \beta)$ where

$$r' = 2^{1/(n-1)} \quad \beta = \frac{4\pi}{(n-1)r'}.$$

This ensures the metric is smooth. The black-hole metric is then given by

$$g_{\mathrm{BH}} = \frac{1}{V} dr^2 + V\, d\theta^2 + r^2 g_{T^{n-2}}, \tag{3.4}$$

where $g_{T^{n-2}}$ is the metric on the torus T^{n-2}, and the function $V(r)$ is defined as

$$V(r) = r^2 - \frac{2}{r^{n-3}}.$$

For $n = 3$, the metric reduces to the standard hyperbolic metric in cylindrical coordinates.

Remark 3.1.1. At $r = r'$, it is easy to check that $V(r) = 0$. This might suggest a singularity in g_{BH}, as the first metric component "blows up." However, this singularity just comes from a poor choice of coordinates. Consider the coordinate transformation

$r = r' + \frac{1}{2}s^2$. Under this change, the metric takes the form:

$$g_{\mathrm{BH}} = \frac{s^2}{V\left(r' + \frac{1}{2}s^2\right)}ds^2 + s^2\frac{V\left(r' + \frac{1}{2}s^2\right)}{s^2}d\theta^2 + \left(r' + \frac{1}{2}s^2\right)^2 (dx_3^2 + \cdots + dx_n^2).$$

Since

$$V\left(r' + \frac{1}{2}s^2\right)s^2 \to \frac{1}{2(n-1)}r'$$

as $s \to 0$, we conclude that g_{BH} is actually smooth at the origin.

Black-hole metric

As the name suggests, this metric originates from Mathematical Physics, where it describes the metric of an anti-de Sitter black-hole [Kra+80]. These black-holes are solutions to the Einstein equations with negative cosmological constant that, far from the black-hole, approach anti-de Sitter spacetime. This space has negative curvature, rather than the usual flat Minkowski spacetime. The general formula is given by

$$V_m(r) = r^2 - \frac{2m}{r^{n-3}},$$

where m is related to the black-hole's mass. The core $(n-2)$-torus, $H = \{r = r_+\} \subset D^2 \times T^{n-2}$, represents the black-hole's horizon.

We are interested in a metric that asymptotically approaches the hyperbolic cusp metric (so that the difference between the metrics in the gluing region is small) and the black-hole metric indeed satisfies this requirement (as $r \to \infty, \quad V(r) \to r^2$). Nevertheless, this metric has infinite volume, so we don't get a geometric convergence towards the cusp. In section 3.1.2 we will see that this can be resolved by considering a twisted version of the metric.

Lemma 3.1.2. *g_{BH} is Einstein.*

Proof. The proof of this Lemma follows the same procedure as Lemma 3.1.1. □

Gluing procedure (Dehn Filling)

We are interested in isometrically mapping $\partial\hat{N}_k$ to the torus T_k that bounds the cusp N_k. We will now describe how $\hat{N}_k$ is constructed to make this identification possible.

Let N_k be bounded by tori $T_k = \partial N_k$, where the injectivity radius satisfies $\mathrm{inj} = \epsilon$ with ϵ being the Margulis constant from definition 2. Choose a simple closed geodesic $\sigma_k \subset T_k$ which will serve as the boundary of the disk D that would be glued. As $V(r)$ is invertible, there exists $R_k > 0$ such that $V(R_k) = l(\sigma_k)/\beta$, where l is the length with respect g_{hyp}.

This ensures that at $r = R_k$, the length of the circle of the toral black-hole metric coincides with $l(\sigma_k)$.

Lemma 3.1.3. *Denote by $M_{BH}(r \leq R_k)$ the part of (M_{BH}, g_{BH}) on which $r \leq R_k$. Then there is a group of isometries Γ of M_{BH} and a number $R_k > 0$ such that*

$$\hat{N}_k = M_{BH}(r \leq R_k)/\Gamma \cong D^2 \times T^{n-2}$$

and

$$\partial \hat{N}_k = M_{BH}(r = R_k)/\Gamma \cong T^{n-1}.$$

Proof. We take this Lemma for granted, as a detailed proof can be found in Section 3 of [And06]. The main idea is to construct a twisted black-hole metric by lifting the flat metric on T^{n-2} to its universal cover $\mathbb{R}^{n-2}$, and then considering the quotient

$$(D^2 \times \mathbb{R}^{n-2})/\mathbb{Z}^{n-2} \cong D^2 \times T^{n-2},$$

where $\mathbb{Z}^{n-2}$ acts freely and with (twisted) black-hole metric

$$g_{BH} = [V^{-1}\, dr^2 + V\, d\theta^2 + r^2 g_{\mathbb{R}^{n-2}}]/\mathbb{Z}^{n-2}. \tag{3.5}$$

□

Remark 3.1.2. Since the twisted metric 3.5 locally agrees with 3.4, it is also Einstein and asymptotically approximates the hyperbolic cusp metric.

Notation: From now on, whenever we write g_{BH}, we will be referring to this *twisted* black-hole metric.

By construction, there exists an isometry $\phi : \partial(D^2 \times T^{n-2}) \to T^{n-1}_k$ that sends $S_1 = \partial D^2$ to the geodesic σ_k. This yields the manifold M_σ obtained by the topological sum:

$$M_{\sigma_k} = M_{core} \cup_\phi (D^2 \times T^{n-2}).$$

The topology of this manifold depends only on the homotopy class of σ_k. This process can be carried out independently to all cusps N_k, $1 \leq k \leq p$ to obtain the manifold M_σ associated with the collection $\sigma = (\sigma_1, ..., \sigma_p)$. Defining $R_{\min} := \min R_k$ we observe $R_{\min} \to \infty$ as $\ell_{\min} \to \infty$ (since $V(r)$ is an increasing function for $r > 0$). Figure 3.2 illustrates this gluing procedure.

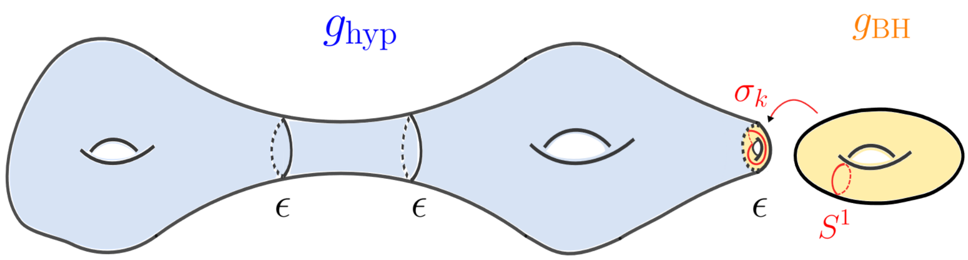


Figure 3.2: A solid torus with the black-hole metric is isometrically glued to the torus boundary.

Many interesting results have been obtained regarding the geometry of M_σ. For instance, Gromov and Thurston proved that if $l(\sigma) > 2\pi$, then M_σ admits a complete metric of non-positive sectional curvature and there are only finitely many isotopy classes for which M_σ is not hyperbolic (the original result was proved in dim 3, but was later on generalized to any dimension [BH96]). These isotopy classes are known as *exceptional slopes*, and considerable effort has been put on finding an upper bound on their number [LM13]. Around 20 years ago, Ian Agol and Marc Lackenby improved these results by reducing the exceptional slope length bound from 2π to 6 [Lac00].

3.1.3 Metric on the interpolation region

To obtain a smooth metric in M_σ take a cut off function $\chi(r) \geq 0$ such that

$$\chi(r) = \begin{cases} 0, & \text{if } r \leq 1, \\ 1, & \text{if } r > 2. \end{cases}$$

and define the approximate metric $\bar{g}$ as

$$\bar{g} = \chi\, g_{\text{hyp}} + (1-\chi)\, g_{\text{BH}}, \tag{3.6}$$

In the tabular neighbourhood of radius 1 around the torus T_k that bounds N_k (i.e $I_k = \bigcap_{k=1}^{S} B_1 T_k \cap N_{b_k}$), there is an interpolation between the metrics g_{hyp} and g_{BH}. As $r \to \infty$, we have $V(r) \sim r^2 + O(R_k^{1-n})$, which leads to the following relationship between the metrics:

$$g_{\text{BH}} - g_{\text{hyp}} = O\left(R_k^{1-n}\right) \quad \text{and} \quad |\nabla^m (g_{BH} - g_{\text{hyp}})| = O\left(R_k^{1-n}\right).$$

To see this, identify the angular coordinate θ in g_{BH} with x_2 in g_{hyp}, and recall that $V(R_k) = l(\sigma_k)/\beta$, as discussed in the previous subsection. From this, we conclude that within the interpolation region I_k

$$\mathrm{Ric}_{\bar{g}} + (n-1)\bar{g} = O\left(R_k^{1-n}\right), \quad \text{and} \quad |\nabla^m \mathrm{Ric}_{\bar{g}}| = O\left(R_k^{1-n}\right), \quad \text{for m} < \infty.$$

In addition, one can prove that the sectional curvatures of g_{BH} are given by

$$\begin{aligned} K_{12} &= -1 + \frac{(n-3)(n-2)}{r^{n-1}}, \\ K_{1i} = K_{2i} &= -1 - \frac{n-3}{r^{n-1}}, \quad i \geq 3, \\ K_{ij} &= -1 + \frac{2}{r^{n-1}}, \quad i,j \geq 3, \end{aligned}$$

which implies that difference between the curvature tensor of g_{BH} and g_{hyp} also decays at a rate of $O(R_{\min}^{1-n})$. Figure 3.3 illustrates the interpolation region (green) between the core of the manifold and the solid torus being glued to the boundary.

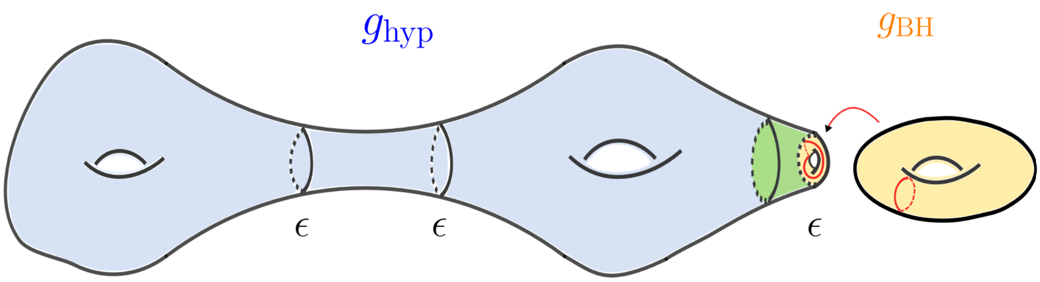


Figure 3.3: Interpolation between g_{BH} and g_{hyp}(green region).

Gluing constructions

Gluing constructions are a common technique in geometric analysis. The basic idea is to build new solutions on a manifold by combining known solutions from two (or more) separate manifolds. The resulting object is typically an approximate solution which is exact away from the gluing regions. This method has been used to obtain different results on minimal surfaces [MP96; Tip13], to construct new manifolds with constant scalar curvature [MPU95], find new extremal Kähler metrics [Tip13], among other applications.

3.2 Einstein operator

Remember that the ultimate goal was to find a metric g on M_σ that solves the Einstein equation:

$$\text{Ric} = \lambda g.$$

However, the Ricci tensor is a diffeomorphism invariant operator and, thus, not elliptic. This problem can be solved by imposing a *gauge condition* that selects a unique representative metric from each orbit of the diffeomorphism group. We then aim to solve the following system of equations

$$\begin{cases} \text{Ric}_g = \lambda g \\ \text{div}_{\bar{g}} g + \frac{1}{2} d\, \text{tr}_{\bar{g}} = 0, \end{cases} \tag{3.7}$$

where the second equation is the *Bianchi gauge* condition and $\bar{g}$ is the approximate metric on M_σ constructed in the previous section. Next, we consider the *Einstein operator* Φ as:

$$\Phi : \mathcal{M}_{m,\alpha} \to \mathcal{S}^2_{m-2,\alpha}$$

$$\Phi_{\bar{g}}(g) := \text{Ric} + (n-1)g + \text{div}^*_g \left(\text{div}_{\bar{g}} g + \frac{1}{2} d\, \text{tr}_{\bar{g}}\, g \right) \tag{3.8}$$

where div^* is the L^2-adjoint of the adjoint operator and $\mathcal{M}_{m,\alpha}$ is the space of $C^{m,\alpha}$ complete Riemannian metrics on a compact manifold M. That is, complete metrics that belong to the Hölder space $C^{m,\alpha}$ in a smooth atlas of $\mathcal{M}$, for $\alpha \in (0,1)$. Similarly, $\mathcal{S}^2_{m,\alpha}$ is the space of $C^{m,\alpha}$ symmetric bilinear forms on M.

A detailed description of the topology of this Hölder space of metrics $\mathcal{M}_{m,\alpha}$ can be found in [And06; Bam12; Biq07]. The key idea is that the $C^{m,\alpha}$ norm is well-defined as long as have a control on both the upper bound of $||\nabla^{m-1}\text{Ric}||_{L^\infty}$ and the lower bound of the injectivity radius. This ensures that the harmonic radius $\rho_{m,\alpha}$ (largest scale at which harmonic coordinates are well-behaved) doesn't deteriorate for any points $x \in \mathcal{M}$. In Section 4, we will maintain good control over the derivatives of the Ricci curvature, but not over the injectivity radius, which will bring some challenges. We will return to this discussion in Section 4. The explicit expression of the norm will not be important for our purposes (but it can be found in [And06], formula 2.20).

There are two main reasons (Lemma 3.2.1 and Lemma 3.2.2) why working with the Einstein operator Φ is particularly nice:

Lemma 3.2.1. *Assume $Ric_g < 0$ and the Bianchi operator $\beta_g(h) := \text{div}_g h + \frac{1}{2} d\, \text{tr}_g h$ is bounded. Then solving for $\Phi_{\bar{g}}(g) = 0$ is equivalent to solving the system of equations 3.7.*

Proof. It is straightforward that the system of equation 3.7 implies that $\Phi_{\bar{g}}(g) = 0$. The other direction is where the difficulty lies. Applying $\beta_{\bar{g}}$ to $\Phi_{\bar{g}}(g)$ we get

$$\beta_{\bar{g}}(\Phi_{\bar{g}}(g)) = \beta_{\bar{g}}\left(\operatorname{div}_g^*\left(\operatorname{div}_{\bar{g}} g + \frac{1}{2} d\operatorname{tr}_{\bar{g}} g\right)\right) = \frac{1}{2}\left(\nabla^*\nabla - \operatorname{Ric}_g\right)(\beta_{\bar{g}}(g)) = 0 \tag{3.9}$$

where we have used the second Bianchi Identity $\beta_{\bar{g}}(\operatorname{Ric}_g) = 0$ (see Theorem 1.85 of ([L.B87])) and the Weitzenböck formula on symmetric bilinear forms presented below. Taking the inner product $\beta_{\bar{g}}$ with respect to g gives

$$\Delta|\beta_{\bar{g}}(g)|^2 - 2\langle \operatorname{Ric}_g(\beta_{\bar{g}}(g)), \beta_{\bar{g}}(g)\rangle \leq 0.$$

As $|\beta_{\bar{g}}(g)| > 0$, we can apply the maximum principle to conclude that $\beta_{\bar{g}}(g) = 0$, which proves the result. □

Weitzenböck formula

Let Δ be the Laplacian acting on symmetric billinear forms. The Weitzenböck formula on symmetric bilinear forms is

$$\nabla^*\nabla h = \Delta h + R(h) - h \circ \operatorname{Ric},$$

where where $R(h)$ is the action of the curvature tensor of $\bar{g}$ on symmetric bilinear forms (i.e $R(h)(X,Y) = \operatorname{tr} h\,(R(\cdot,X)Y,\cdot))$ and Δ is the Hodge Laplacian associated with the Levi-cevita connection. The proof of this expression can be found in ([L.B87]) Section 12.69.

Lemma 3.2.2. *Let $\bar{g}$ be an Einstein metric. The linearization of the operator around $\bar{g}$ has the following form:*

$$L_{\bar{g}} h = (\operatorname{div}^* \operatorname{div} + d^* d)h - R(h) + (n-1)h. \tag{3.10}$$

Proof. The details of this Lemma are out of scope of this dissertation, but the idea is to express the linearization of the Ricci tensor in terms of the Linchnerowicz Laplacian and then use the Weitzenböck formula to obtain the final expression. □

In addition, if $\bar{g}$ is hyperbolic then $R(h) = h - (\operatorname{tr} h)_{\bar{g}}\bar{g}$ and the expression simplifies as:

$$L_{\bar{g}} h = (\operatorname{div}^* \operatorname{div} + d^* d)h + (\operatorname{tr} h)_{\bar{g}}\bar{g} + (n-2)h = -\Delta h - 2h - 2(\operatorname{tr} h)_{\bar{g}}\bar{g} \tag{3.11}$$

In Chapter 4, we will be interested in proving that the linearized Einstein L is invertible and with uniform bound on L^{-1}. The following Lemma will prove very useful to do that.

Lemma 3.2.3. *Let h and f be $(0,2)$-tensors that only depend on r. In addition, assume h is T^{n-1} invariant and expressed as*

$$h = \sum h_{ab}(r)\theta^a \cdot \theta^b,$$

where $h_{ab}(r)$ is a function of r only and θ is the local orthonormal coframing of the cusp metric g_{hyp}, dual to e^i, for $e_1 = \nabla s$, where $ds = r^{-1}dr$, and e^i are tangent to T^{n-1} for $i \geq 2$. Then the PDE

$$L_{g_{hyp}} h = 0$$

can be transformed to the following linear system of ODEs with constant coefficients

$$\begin{cases} r^2 h''_{1b} + nrh'_{1b} - nh_{1b} = 0, & \textit{(Equation 1)} \\ r^2 h_{11} + nrh'_{11} - 2(n-1)h_{11} = 0, & \textit{(Equation 2)} \\ r^2 (h_{ab})'' + nr(h_{ab})' - 2\delta_{ab}\sum_{k=2}^{n} h_{kk} = 0. & \textit{(Equation 3)} \end{cases} \tag{3.12}$$

Proof. From equation 3.11, we have that the equation $L_{g_{hyp}}(h) = 0$ is equivalent to $\Delta h + 2h - 2(\text{tr } h)g_{hyp} = 0$. The derivation of the ODEs now follows from explicitly expanding Δh. The details can be found in Appendix A of [And06]. By integrating, we get the following solutions to the system of ODEs:

$$\begin{cases} h_{1b} = c_1 r + c_2 r^{-n}. & \textit{(Solution 1)} \\ h_{11} = c_1 r^{\alpha_+} + c_2 r^{\alpha_-}, & \textit{(Solution 2)} \\ h_{ab} = c_1 r^{-(n-1)} + c_2 \quad \text{for } a \neq b & \textit{(Solution 3)} \end{cases} \tag{3.13}$$

where $\alpha_\pm = \frac{1}{2}\left(-(n-1) \pm \sqrt{(n-1)^2 + 8(n-1)}\right)$ and $c_1, c_2 \in \mathbb{R}$. □

It is important to note that we are interested in the *fundamental* solution of this system of ODEs and their growth rates, rather than the *exact* solution. These growth estimates on h would be crucial for the argument in the next section.

Remark 3.2.1. Note that $L_{\bar{g}}(h)$ is an elliptic and L^2 self-adjoint operator inherited from the Laplacian Δ. In addition, by standard elliptic regularity theorems (see Theorem 40 [L.B87]), h is in $C^{m,\alpha}$.

Let's summarize the key results from Chapter 3. We have constructed a family of metrics $\bar{g}$ defined on (distinct) compact manifolds M_σ. The curvature of $\bar{g}$ is uniformly bounded by the curvature of g_{BH} as $-1 + O(R_{\min}^{1-n})$. Moreover, outside the interpolation regions I_k (which have radius 1 around each torus T_k), the metric $\bar{g}$ satisfies

$$\mathrm{Ric}_{\bar{g}} + (n-1)\bar{g} = 0, \quad \text{so} \quad \Phi_{\bar{g}}(\bar{g}) = 0,$$

while on each I_k, we have

$$\mathrm{Ric}_{\bar{g}} + (n-1)\bar{g} = O\left(R_k^{1-n}\right), \quad \text{so} \quad \Phi_{\bar{g}}(\bar{g}) = O\left(R_k^{1-n}\right).$$

Hence, $\bar{g}$ gets closer and closer to being Einstein as $R_k \to \infty$.

Chapter 4

Perturbation to an Einstein metric

4.1 Heuristics of the proof

The goal of this chapter is to prove that for sufficiently large σ, there exists a small perturbation h of $\bar{g}$ that is Einstein on M_σ. As stated in Theorem 1, this perturbation is unique (up to diffeomorphism) within a *small* neighbourhood of $\bar{g}$. The precise construction of this neighbourhood will be described in detail in this chapter.

We begin by presenting the heuristics of the proof, which will serve as our starting point, and explaining why this approach fails in its simplest form. The following theorem is an adapted version of Theorem 3.6.6 from [Biq07].

Theorem 3. *Let $h \in \mathcal{S}^2_{m-2,\alpha}$, and let $\bar{g}$ be the approximate Einstein metric constructed in Chapter 3. The Einstein operator $\Phi : \mathcal{M}_{m,\alpha} \to \mathcal{S}^2_{m-2,\alpha}$ can be expanded as*

$$\Phi_{\bar{g}}(\bar{g} + h) = \Phi_{\bar{g}}(\bar{g}) + L_{\bar{g}}(h) + Q_{\bar{g}}(h),$$

where $L_{\bar{g}}$ is the linearization at $\bar{g}$ introduced in formula 3.10, and $Q_{\bar{g}}(h)$ is a quadratic remainder satisfying

$$\|Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\|_{C^{m-2,\alpha}} \leq k(\|x\|_{C^{m,\alpha}} + \|y\|_{C^{m,\alpha}})\|x - y\|_{C^{m,\alpha}}, \tag{4.1}$$

for all x, y with $\|x\|_{C^{m,\alpha}}, \|y\|_{C^{m,\alpha}} \leq r_0$. Assume $L_{\bar{g}}$ is invertible and

$$\|L_{\bar{g}}^{-1}\|_{C^{m,\alpha}} \leq \Lambda. \tag{4.2}$$

If $r \leq r_0$ satisfies

$$r \leq \frac{1}{2\Lambda k}, \quad \textit{and} \quad \|\Phi_{\bar{g}}(\bar{g})\|_{C^{m-2,\alpha}} \leq \frac{r}{\Lambda} - kr^2, \tag{4.3}$$

then there exists a unique solution h to the equation $\Phi_{\bar{g}}(\bar{g}+h) = 0$ such that $\|\bar{g}+h\|_{C^{m,\alpha}} \leq r$.

Proof. The result follows directly from the Banach Fixed Point Theorem. Consider the equation

$$\Phi_{\bar{g}}(\bar{g}+h) = 0.$$

Rearranging, we obtain

$$L_{\bar{g}}(h) = -\Phi_{\bar{g}}(\bar{g}) - Q_{\bar{g}}(h).$$

Applying $L_{\bar{g}}^{-1}$ to both sides gives

$$h = -L_{\bar{g}}^{-1}\big(\Phi_{\bar{g}}(\bar{g}) + Q_{\bar{g}}(h)\big).$$

Define the map

$$\Psi(h) := -L_{\bar{g}}^{-1}\big(\Phi_{\bar{g}}(\bar{g}) + Q_{\bar{g}}(h)\big).$$

Then, solving $\Phi_{\bar{g}}(\bar{g}+h) = 0$ is equivalent to finding a fixed point of Ψ. We now need to verify that Ψ maps the closed ball $B_r(0) \subset C^{m,\alpha}$ into itself and is a contraction. First, for any $h \in B_r(0)$, we have:

$$\begin{aligned}
\|\Psi(h)\|_{C^{m,\alpha}} &= \left\|L_{\bar{g}}^{-1}\left(\Phi_{\bar{g}}(\bar{g}) + Q_{\bar{g}}(h)\right)\right\|_{C^{m,\alpha}} \\
&\leq \|L_{\bar{g}}^{-1}\|_{C^{m,\alpha}}\left(\|\Phi_{\bar{g}}(\bar{g})\|_{C^{m-2,\alpha}} + \|Q_{\bar{g}}(h)\|_{C^{m-2,\alpha}}\right) \\
&\leq \Lambda\left(\|\Phi_{\bar{g}}(\bar{g})\|_{C^{m-2,\alpha}} + k\|h\|^2_{C^{m,\alpha}}\right) \\
&\leq r - \Lambda k r^2 + \Lambda k r^2 \\
&\leq r
\end{aligned}$$

Thus, Ψ maps $B_r(0)$ into itself. For $x, y \in B_r(0)$, we compute:

$$\begin{aligned}
\|\Psi(x) - \Psi(y)\|_{C^{m,\alpha}} &= \left\|L_{\bar{g}}^{-1}\left(Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\right)\right\|_{C^{m,\alpha}} \\
&\leq \|L_{\bar{g}}^{-1}\|_{C^{m,\alpha}}\|Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\|_{C^{m-2,\alpha}} \\
&\leq 2\Lambda k r\|x-y\|_{C^{m,\alpha}},
\end{aligned}$$

where we have used the Lipschitz condition (4.1) for $Q_{\bar{g}}$ on $B_r(0)$. If $r \leq \frac{1}{2\Lambda k}$, then Ψ is a contraction. This is precisely condition (4.3). Thus, the Banach Fixed Point Theorem guarantees the existence of a unique fixed point and, hence, of a solution of $\Phi_{\bar{g}}(\bar{g}+h) = 0$. □

We see that there are three conditions that need to be satisfied to be able to apply the Theorem 3 (Conditions 4.1, 4.2 and 4.3). In Chapter 3 we have proved that $\|\Phi_{\bar{g}}(\bar{g})\|_{C^{m,\alpha}} \leq CR_{min}^{1-n}$ which can be made arbitrarily small by increasing $R_{\min}$, so Condition 4.3 is satisfied. Lemma 4.1.1 proves Condition 4.1.

Lemma 4.1.1. *Let $\Phi_{\bar{g}}(\bar{g}+h) = \Phi_{\bar{g}}(\bar{g}) + L_{\bar{g}}(h) + Q_{\bar{g}}(h)$ as of Theorem 3. Then there exist* *k such that*

$$\|Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\|_{C^{m-2,\alpha}} \leq k(\|x\|_{C^{m,\alpha}} + \|y\|_{C^{m,\alpha}})\|x - y\|_{C^{m,\alpha}} \quad \text{for } \|x\| \leq r_0, \|y\| \leq r_0,$$

for sufficiently small r_0.

Proof. It is easy to see that $Q_{\bar{g}}(h)$ is a nonlinear explicit expression with zero linearisation at 0 (the details can be found in [Biq07], section 3.7). Thus, there exists a constant k such that $\|DQ_{\bar{g}}(h)\|_{C^{m-2,\alpha}} \leq k\|h\|_{C^{m-2,\alpha}}$ for $\|h\|_{C^{m,\alpha}} \leq r_0$. Applying the mean value theorem in Banach spaces, we have

$$Q_{\bar{g}}(x) - Q_{\bar{g}}(y) = \int_0^1 DQ_{\bar{g}}(y + t(x-y))(x-y)\, dt$$

and

$$\|Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\|_{C^{m-2,\alpha}} \leq \int_0^1 k\|y + t(x-y)\|_{C^{m,\alpha}}\|x - y\|_{C^{m,\alpha}}\, dt.$$

Using the fact that $\|y + t(x-y)\|_{C^{m,\alpha}} \leq \|x\|_{C^{m,\alpha}} + \|y\|_{C^{m,\alpha}}$, we obtain

$$\|Q_{\bar{g}}(x) - Q_{\bar{g}}(y)\|_{C^{m-2,\alpha}} \leq k(\|x\|_{C^{m,\alpha}} + \|y\|_{C^{m,\alpha}})\|x - y\|_{C^{m,\alpha}}.$$

□

The problem arises from the uniform bound on L^{-1} of Condition 4.2 which *doesn't hold* between the Hölder spaces we are currently working on. In Subsection 4.2 we will discuss how [And06] addresses this issue by considering a subspace of the space of solutions, which is explicitly constructed to ensure a uniform bound. In Subsection 4.3, we describe an alternative approach by [Bam12] using weighted Hölder spaces. However, this would require to reprove Conditions 4.1 and 4.3 using the new norm.

4.2 Hölder Spaces - Non-uniform bound on L^{-1}

4.2.1 Proof of the non-uniform bound

We now state the main Theorem of [And06].

Theorem 4. *Let M_σ be the manifold constructed by Dehn Filling, as described in Chapter 3. For σ sufficiently large, there is a constant Λ such that*

$$\|h\|_{C^{m,\alpha}} \leq \Lambda(\log R_{\max})\|L_{\bar{g}}(h)\|_{C^{m-2,\alpha}}. \tag{4.4}$$

Then $L_{\bar{g}}$ is invertible and the norm of $L_{\bar{g}}^{-1} : S_2^{m-2,\alpha} \to S_2^{m,\alpha}$ is uniformly bounded by $\Lambda \log R_{\max}$, where $R_{\max} = \max_k R_k$. In addition, there exists $\epsilon > 0$ such that (4.4) holds, for all metrics $g' \in B_{\bar{g}}(\epsilon)$, where $\bar{g}$ is the approximate metric on M_σ constructed in Chapter 3.

Proof. As discussed in Chapter 3, $L_{\bar{g}}$ is an elliptic operator so we are able to apply standard Schauder estimates to get

$$\|h\|_{C^{m,\alpha}} \leq \Lambda \left(\|L_{\bar{g}}(h)\|_{C^{m-2,\alpha}} + \|h\|_{L^\infty} \right).$$

Hence, to prove (4.4) is enough to show that

$$\|h\|_{L^\infty} \leq \Lambda (\log R_{\max}) \|L_{\bar{g}}(h)\|_{C^{m-2,\alpha}}.$$

The proof will go by contradiction. Assuming inequality 4.4 is false, we can construct a sequence of Dehn-filled manifolds M_{σ^i}, with approximate solutions $\bar{g}^i$, symmetric bilinear forms $h^i \in S_2^{m,\alpha}(M_\sigma^i)$ and $l(\sigma_k)^i \to \infty$ for each $(\sigma_k)^i \in \sigma^i$ (remember that k is the index of the filled cusp N_k for $1 \leq k \leq p$, where p is the number of cusps), such that

$$\|h^i\|_{L^\infty} = 1, \quad \text{but} \quad \log(R_{\max})^i \|L^i(h^i)\|_{C^{m-2,\alpha}} \to 0. \tag{4.5}$$

Given a fixed point $y \in M_{\text{core}}$ and sequence of basepoints $(x^i) \in (M_\sigma^i, g^i)$, we then have one of the following scenarios:

1. **Hyperbolic case:** $\text{dist}_{\bar{g}}(x^i, y) < \infty$ and the pointed sequence $(M_\sigma^i, \bar{g}^i, x^i)$ converges in the pointed Gromov-Hausdorff topology to the limit $(M_{\text{core}}, g_{hyp}, x)$, where $x = \lim x^i$.

2. **Cusp case:** $\text{dist}_{\bar{g}}(x^i, (T_k^{n-2})^i) \to \infty$ and $\text{dist}_{\bar{g}}(x^i, y_0) \to \infty$ for all k, where T_{n-2}^k is the core torus of the Dehn filling on N_k. The pointed sequence $(M_\sigma^i, \bar{g}^i, x^i)$ converges in the pointed Gromov-Hausdorff topology to the hyperbolic metric g_{hyp}. We don't have a control on the injectivity radius along the cusp, which will cause the $C^{m,\alpha}$ norm to degenerate (see Section 3.2). Nevertheless, we will see that this can be remediated by working on a larger covering space.

3. **Black-hole case:** $\text{dist}_{\bar{g}}(x^i, (T_k^{n-2})^i) < \infty$ and the pointed sequence $(M_\sigma^i, \bar{g}^i, x^i)$ converges in the pointed Gromov-Hausdorff topology to the black-hole metric g_{BH}. In this case, the $C^{m,\alpha}$ norm degenerates too as the injectivity radius gets arbitrarily small, so we will also have to work on a covering space.

We will now analyse these limits case by case:

1. **Hyperbolic case:** Let $(M_{\text{hyp}})_{r_0}$ be the manifold obtained from M_{hyp} by truncating its cusps at distance r_0 from the basepoint y (remember that M_{hyp} was the original

manifold with cusps endowed with g_{hyp}, refer to Section 3.1). Using the explicit expression for the linearized Einstein operator on hyperbolic metrics (formula 3.11) and applying Stoke's theorem, we get

$$\int_{(M_{\text{hyp}})_{r_0}} |\mathrm{d}h|^2 + |\delta h|^2 + (n-2)|h|^2 + (\operatorname{tr} h)^2 = \int_{\partial (M_{\text{hyp}})_{r_0}} Q(h, \partial h).$$

Note that the right hand side goes to 0 as $r_0 \to \infty$ and, by our assumption 4.5 $Q(h, \partial h)$ is uniformly bounded, so $h = 0$ on M_{hyp}, leading to a contradiction of 4.5.

By the smooth convergence of h^i to the limit form h, it follows that $h^i(x^i) \to 0$ for the sequence of basepoints x^i satisfying Case 1. We can then consider a nested sequence of compact subsets K_j that covers M_{hyp} and such that

$$|h_{i_j}(x)| = \epsilon_j \quad \forall x \in K_j \tag{4.6}$$

and $\epsilon_j \to 0$. Since h^i is vanishing in larger and larger compact subset of M_{hyp}, this means that the support must be either in the cusp region or in the black-hole region. This argument is important as it tell us not only that $h \to 0$ in this region, but that the support of h^i is moving towards the other two regions.

2. **Cusp case:** We will first discuss the collapse of the norm, and how to resolve it. When we introduced the space of Hölder metrics $C^{m,\alpha}$, we already discussed that the norm is only well-defined as long as we have control on the upper bound of $||\nabla^{m-1}\mathrm{Ric}||_{L^\infty}$ and a lower bound on the injectivity radius (refer to Subsection 3.2). We have a good control on the first one, but as the injectivity radius shrinks along the cusp, the $C^{m,\alpha}$ norm degenerates. However, T.Anderson argues in [And06] that we can avoid this degeneration by working on a finite covering space with the following properties:

   - The lifted metric remains locally isometric to the original.
   - The lifted injectivity radius, i_0, is large enough to ensure the harmonic radius is well-defined. The choice of i_0 depends on dimension and is often taken as a small multiple of the Margulis constant.
   - The degree of the cover depends on the local injectivity radius, with smaller injectivity radii requiring larger covers.

   The lifted forms h^i are invariant under the corresponding group of covering transformations, which converges to T^{n-1}. We can now apply Lemma 3.2.3 and analyse the solution of $L_{g_{hyp}}(h) = 0$ via the linear system of ODEs 3.12. As we are assuming that h is bounded, we can rule out some of the fundamental solutions and conclude that $h_{1b} = 0$ for any b. Thus, to solve $L_{\bar{g}}(h) = 0$ we only need to consider equation

3 from the system of ODEs 3.12,

$$r^2 h''^i_{ab} + nrh^i_{ab} = 0, \tag{4.7}$$

which has general solution $h_{ab} = c_1 r^{-(n-1)} + c_2$. However, since h^i_{ab} is bounded along the cusp, it follows that

$$h^i_{ab} = c \tag{4.8}$$

for some constant $c \in \mathbb{R}$. Geometrically, this means that all deformations of the cusp metric originate from deformations of the flat structure on T^{n-1}. These bounded solutions (referred to as *Einstein trivial variations*) will be the focus of the next section. At first glance, these types of deformations may seem problematic, as they do not decay unless $c = 0$. However, the next lemma shows that this is the case whenever $\log(R_{\max})^i \, \|L^i(h^i)\|_{C^{m-2,\alpha}} \to 0$.

Lemma 4.2.1. *Let $h \in S_2^{m,\alpha}$ on $\mathbb{R}^2 \times \mathbb{R}^{n-2}$ with the black-hole metric and satisfying*

$$\log(R_{\max})^i \|L^i(h^i)\|_{C^{m-2,\alpha}} \to 0$$

defined in the sequence of Manifolds (M^i, g^i, x^i) such that $dist_{\bar{g}}(x^i, (T_k^{n-2})^i) \to \infty$ and $dist_{\bar{g}}(x^i, y_0) \to \infty$ (i.e Cusp case). Then $h^i \to 0$.

Proof. As before, the injectivity radius of the tori $T_{n-1}(r)$ shrinks as $r \to \infty$, so we need to unwrap this collapse by passing to larger covering space. To be able to use the linear system of ODEs from Lemma 3.2.3, h needs to depend only on r. Unfortunately, this is not the case here as the lifted forms $h^i_{ab}(r, \theta)$ and $f^i_{ab}(r, \theta)$ also depend on the parameter θ. However, this can be remedied easily by considering their average over the torus:

$$h^i_{ab}(r) = \frac{1}{\mathrm{vol}(T_{n-1}(r))} \int_{T_{n-1}(r)} h^i_{ab}(r, \theta)\, d\theta.$$

This averaging operator commutes with L, so we now have an equation $L(h^i_{ab}(r)) = f^i_{ab}(r)$ which only depends on r and which is T^{n-1} invariant. In addition, the difference between $h^i_{ab}(r, \theta)$ and its average $h^i_{ab}(r)$ on a non-trivial tubular neighbourhood around $T^{n-1}(r)$ decays as r/R^i_k, and the black-hole metric g_{BH} differs from the hyperbolic metric g_{hyp} on the order of $O(r^{-(n-1)})$ (refer to Section 3.1.3). It then follows that ODE 4.7 may be written as

$$r^2 (h^i_{ab}(r))'' + nr(h^i_{ab}(r))' = f^i_{ab}(r) + e^i_{ab}(r), \tag{4.9}$$

where

$$e^i_{ab}(r) = O(r/R_{k,i}) + O(r^{-(n-1)})$$

is the error term due to the averaging and the difference of the metrics discussed just above. Expression 4.9 simplifies the analysis significantly, as we have transformed a partial differential equation into an ordinary differential equation that depends only on the radial coordinate. By integrating expression 4.9 explicitly (using Integrating Factor method, with integration factor $\mu(r) = r^n$) between a fixed arbitrary large constant C_0 and r we get:

$$h'^{i}_{ab}(r) = \frac{1}{r^n}\left(\int_{C_0}^{r} t^{n-2}(f^i_{ab}(t) + e^i_{ab}(t))\, dt + c\right).$$

Taking the absolute value in both sides,

$$|h'^{i}_{ab}(r)| \leq \frac{1}{r^n}\left(\underbrace{\int_{C_0}^{r} \left|t^{n-2} f^i_{ab}(t)\right| dt}_{A} + \underbrace{\int_{C_0}^{r} \left|t^{n-2} e^i_{ab}(t)\right| dt}_{B}\right). \tag{4.10}$$

For clarity, we will integrate both terms A and B separately. On term A, we have

$$\frac{1}{r^n}\left(\int_{C_0}^{r} \left|t^{n-2} f^i_{ab}(t)\right| dt\right) \leq \frac{\sup f^i_{ab}(r)}{r^n}\int_{C_0}^{r} t^{n-2}\, dt \leq \frac{\sup f^i_{ab}(r)}{r^n}\left(\frac{r^{n-1}}{n-1} + \frac{C_0^{n-1}}{n-1}\right),$$

and term B

$$\frac{1}{r^n}\left(\int_{C_0}^{r} \left|t^{n-2} O\Big(\frac{t}{R^i_k}\Big) + O(t^{-(n-1)})\right| dt\right) \leq O\left(\frac{1}{nR^i_k}\right) + O\left(\frac{\log\, r}{r^n}\right).$$

For large enough r, we then get

$$|h'^{i}_{ab}(r)| \leq C\left(\frac{\sup f^i_{ab}(r)}{r} + \frac{1}{R^i_k} + \frac{\log\, r}{r^n}\right).$$

Integrating again from r to r^i we get

$$|h^i_{ab}(r) - h^i_{ab}(r^i)| \leq C\left(\sup f^i_{ab}(r^i)\log\, r^i + \frac{r^i}{R^i_k} + (r^i)^{-(n-1)}\log\, r^i + r^{-(n-1)}\log\, r\right). \tag{4.11}$$

Let $\delta^i = \sup f^i_{ab}(r^i)\log(R^i_k)$, then

$$|h^i_{ab}(r)| \leq C\delta^i\frac{\log r^i}{\log R^i_k} + Cr^{-(n-1)}\log r + |h^i_{ab}(r^i)| \leq Cr^{-(n-1)}\log r \tag{4.12}$$

uniformly on $[C_0, r^i]$, as $\delta^i \to 0$ as $i \to \infty$ by the assumption 4.5 and $|h^i_{ab}(r^i)| \to 0$ as $i \to \infty$ by 4.6. Then, as $r \to \infty$ we have that $|h^i_{ab}| \to 0$, concluding the prof of Lemma 4.2.1. □

As the black-hole metric asymptotically approaches the hyperbolic, we get directly a contradiction of 4.5.

It is important to emphasize that it is *not* true that $h^i_{ab} \to 0$ as $\sup f^i_{ab} \to 0$, precisely because of the bounded fundamental solutions (Einstein trivial variations) that don't decay. The dependence on $\log R_{\max}$ is thus crucial.

3. **Black-hole case:** We have done most of the work on the analysis of this case. Summarizing the results we got so far, we have that by assumption 4.5 and by 4.12

$$||h||_{L^\infty} = 1 \quad \text{and} \quad |h| \leq C_0 r^{-(n-1)} \log r$$

as $r \to \infty$ in $(D^2 \times T^{n-2}, g_{BH})$. The authors in [And06] then proceed to prove that this situation is impossible. The prove goes by solving the system of ODEs 4.7, again. The details are thus not included as the argument is very similar to the one already discussed.

To complete the proof of Proposition 4.4 and conclude that $L_{\bar{g}}$ is invertible, we need to address the surjectivity of $L_{\bar{g}}$. To do this, we will use the theory of Fredholm operators.

Definition 6 (Fredholm Operator)**.** *A bounded linear map $L : H_1 \to H_2$ between Banach spaces is called* Fredholm *if* $\ker L$ *and* $\operatorname{coker} L$ *are finite-dimensional and* $\operatorname{Im} L$ *is closed.*

Lemma 4.2.2. *Let L be the linearized Einstein operator satisfying the bound 4.4. Then $L : C^{m,\alpha} \to C^{m-2,\alpha}$ is surjective.*

Proof. L is an elliptic and L^2 self-adjoint operator (see Remark 3.2.1). From Proposition 4.4, we have that $\mathrm{Ker} L = \{0\}$. By elliptic regularity (see Theorem 40 [L.B87]), we have that cokernel $L \cong \mathrm{Ker}\ L^* = \mathrm{Ker} L = \{0\}$, so the cokernel is also trivial. It thus remains to show that the image of L is closed in $C^{m-2,\alpha}$. Take a Cauchy sequence (Lh_n) in $C^{m-2,\alpha}$. Then (h_n) is a Cauchy sequence in $C^{m,\alpha}$ and, by completeness of Banach spaces, $(h_n) \to h$. Thus, $(Lh_n) \to Lh$ in $C^{m-2,\alpha}$. We can now just apply the Fredholm alternative to conclude that L is surjective. (The proof of the Fredholm alternative can be found in any functional analysis textbook, see for example Chapter 4 of [Cla21]).

□

The last statement of Theorem 4 regarding a metric $g' \in B_{\bar{g}}(\epsilon)$ follows directly by considering a sequence $g^{'i} \in B_{\bar{g}^i}(\epsilon_i)$ with $\epsilon_i \to 0$. Applying the exact same argument to this sequence, we arrive to the same contradictions of 4.5.

This concludes the proof of Theorem 4. □

4.2.2 Proof of Theorem 1

Now that we have proved Theorem 4, we want to prove that there is a small perturbation h of $\bar{g}$ that is Einstein on M_σ. This will conclude the proof of Theorem 1. Unfortunately, we can not apply the Banach fixed point argument described in Theorem 3 directly because the bound on L^{-1} is not uniform. Remember that the dependency of Λ in $\log R_{\max}$ arose because of the behaviour of f^i_{ab} in expression 4.11, which in turn was due to some bounded solutions (Einstein trivial variations) of the system of ODEs. If there was away to ensure $f^i_{ab} = 0$ on that region, we will obtain a uniform bound. That is exactly the idea behind the proof that T.Anderson proposes [And06], which involves restricting the Einstein operator Φ to a carefully defined submanifold.

Let B be the union of the black-hole regions $(\bigcup_{k=1}^{p} \hat{N}_k)$. Define

$$W = \{f \in S_2^{m-2,\alpha} : f(x) = 0, \forall x \text{ s.t. } \mathrm{dist}_{\bar{g}}(x, B) \geq 2\}. \tag{4.13}$$

W is closed in $S_2^{m-2,\alpha}$ and so is a Banach subspace of $S_2^{m-2,\alpha}$. Remember that $\Phi_{\bar{g}}(\bar{g}) = 0$ outside the interpolation region, and so $f_0 = \Phi_{\bar{g}}(\bar{g}) \in W$. Figure 4.1 illustrates this region W.

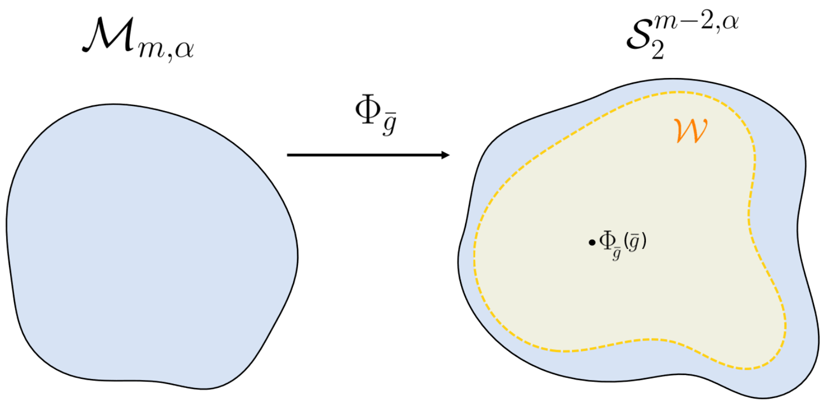


Figure 4.1: Illustration of the subspace W of $S_2^{m-2,\alpha}$

We then let $W_\varepsilon = W \cap B_{f_0}(\varepsilon)$ where $B_{f_0}(\varepsilon)$ is a ε-ball about f_0 in $S_2^{m-2,\alpha}$, and set $U_\varepsilon = \Phi^{-1}(W_\varepsilon)$. We are now interested in the restriction $\Phi_0 = \Phi|_{U_\varepsilon} : U_\varepsilon \to W_\varepsilon$. Note that by Theorem 4 for ε sufficiently small, every point in W_ε is a regular value of Φ_0 (the differential is surjective). Hence, by the *Regular Value Theorem*, U_ε is a Banach submanifold of $M_{m,\alpha}$ and Φ_0 is a local diffeomorphism onto W_ε. Note that ϵ depends on the Dehn Filling σ via $R_{\max}$. Figure 4.2 illustrates these domains.

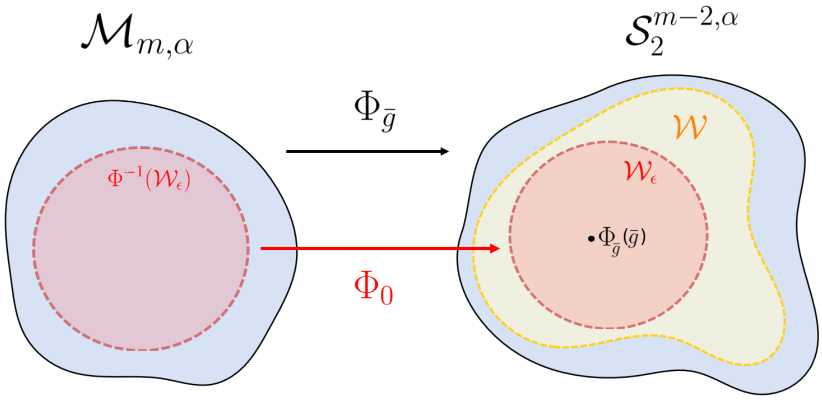


Figure 4.2: Illustration of the domain and codomain of Φ_0

We now work with the restriction Φ_0. The linearization L_0 restricted to the tangent spaces $T_{g'}U_\varepsilon$ of U_ε gives a linear map $L_0(h) = f$ from $h \in T_{g'}U_\varepsilon$ to $f \in T_{\Phi_0(g')}W_\varepsilon$ for any $g' \in U_\epsilon$. But now notice that for any $f \in T_{\Phi_0(g')}$ we have that $f = 0$ outside the region of distance larger than 2 around the black-hole region (refer to the definition 4.13 of W). This means that $f = 0$ on formula 4.11 when analysing case 2 (Cusp region). The entire proof of Theorem 4 holds without any exchanges, except that we have now lost the dependency on $\log(R_{max})$ and we get a uniform bound on the inverse of the linearized Einstein operator L. An important thing to note is that while the support of f is restricted by the definition of W, the support of h is not necessarily restricted. That is, h could be non-zero everywhere on M_σ. This is an interesting observation, as it means that a global perturbation h can still produce a localized function f.

With a uniform bound on the inverse, we are in a very good position to apply the heuristics of Theorem 3. We have a smooth map Φ_0 between Banach spaces with $\|L_0^{-1}\|_{C^{m,\alpha}} \leq \Lambda$. Thus, there is a domain $\Omega \subset U_\varepsilon$ and $\delta > 0$ such that $\Phi_0 : \Omega \to W_\delta$, is a diffeomorphism onto W_δ. Note that $|\Phi_{\bar{g}}(\bar{g})| \leq CR_{min}^{1-n}$, so we can choose $R_{\min}$ large enough to ensure $0 \in W_\delta$. Figure 4.3 illustrates the subspace W_δ.

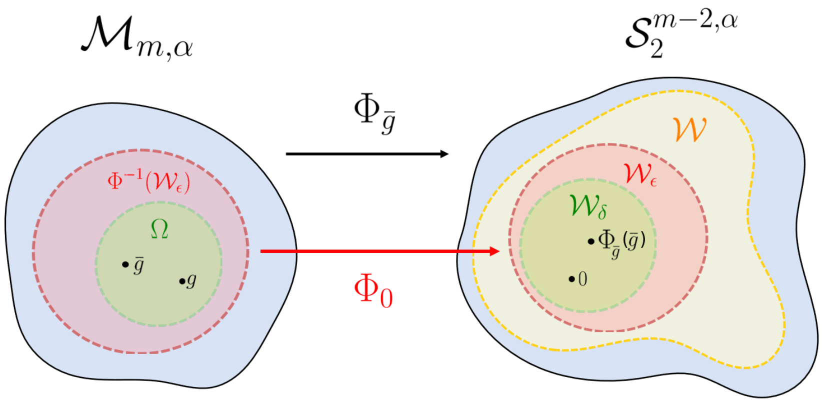


Figure 4.3: Illustration of the subspace W_δ

We already discussed that condition 4.1 and Condition 4.3 are satisfied in Subsection 4.1. We can then apply Theorem 3 to conclude that there exists an h such that

$$\Phi_{\bar{g}}(\bar{g}+h)=0.$$

As states in Remark 3.2.1, this perturbation h is in $C^{m-2,\alpha}$ by elliptic regularity results.

Remark 4.2.1. Note that $\bar{g}+h$ is the *unique* Einstein metric (up to isometry) in Ω. In addition, since Φ is a local diffeomorphism near $\bar{g}+h$ it follows that this metric is locally rigid (an isolated point on the moduli space of Einstein metrics).

Remark 4.2.2. This proof *only* works if $R_{\max}$ and $R_{\min}$ are sufficiently controlled relative to one another: the construction of the submanifold $W_\varepsilon \subset W$ depends logarithmically on $R_{\max}$ via ε, while $R_{\min}$ must be large enough to ensure $0 \in W_\delta \subset W_\epsilon$.

4.3 Weighted Hölder spaces - Uniform bound on L^{-1}

In the previous subsection, we've seen that we are only able to prove the existence of an Einstein metric on $\mathcal{M}_\sigma$ when $R_{\max}$ is sufficiently controlled by $R_{\min}$. The argument from ([Bam12]) is that the bad invertibility of L comes from certain bounded deformations of the cusp, which we have already encountered in the previous chapter. We now provide a more detailed discussion of these types of deformations.

4.3.1 Einstein trivial variations

Definition 7 (Trivial Einstein variations)**.** *Let u_{ij} be a symmetric $(n-1)\times(n-1)$ matrix indexed by $i,j=2,\dots,n$. An Einstein variation h is a symmetric bilinear form $h=r^2u_{ij}\,dx_idx_j$ with tr $u=0$. It is easy to check that then $\Phi_{g_{hyp}}(g_{hyp}+h)=0$ and $L_{g_{hyp}}h=0$.*

Thus, a trivial Einstein variation is an infinitesimal change in the flat structure of the torus that *cannot* be detected by L. This is of course problematic when trying to obtain the uniform bound on L^{-1}. These deformations correspond exactly to the bounded fundamental solutions of the system of ODEs discussed in the previous chapter. (There, we had $h_{ab}=c$ because we were working with an orthonormal frame; the expression looks different now because we are working with a coordinate frame). Therefore, to make the operator L invertible, we could work with a norm that *isolates* these types of variations. These norms will be introduced in Subsection 4.3.2.

The idea is to decompose the symmetric bilinear form h as

$$h = \bar{h} + \sum_{k=1}^{p} \rho_k u_k,$$

where $u_1, \ldots, u_p$ are trivial Einstein variations and ρ_k are suitable cutoff functions. There is, in fact, a canonical way to choose these trivial variations u_k by considering u'_k such that $|(h - u'_k)(c'_k)|$ is minimized, where $r(c_k) = \sqrt{R_k}$. However, these technical details are out of the scope, as they are not essential to the main argument.

4.3.2 Hybrid norms

As discussed, we aim to define a norm that treats the trivial Einstein variations separately. This will be done using a *hybrid norm*, which combines a weighted norm for $\bar{h}$ with an unweighted norm for the u_k.

The weighted Hölder norm is defined as

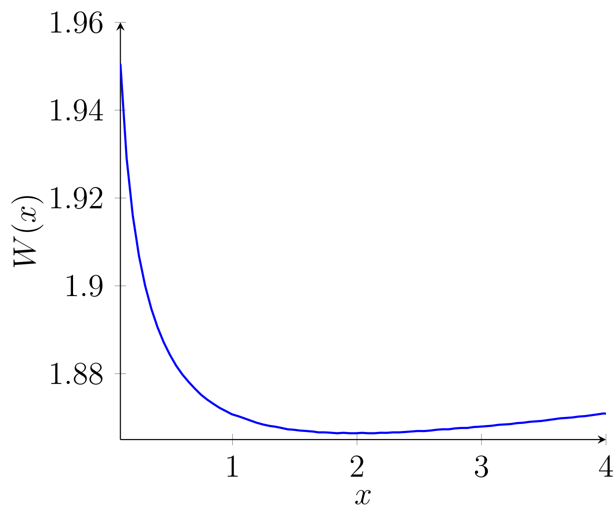
$$\|h\|_{m,\alpha;*} := \|W^{-1}h\|_{m,\alpha},$$

where

$$W(x) = \begin{cases} \left(\frac{r}{R_k}\right)^{0.1} + r^{-0.1}, & \text{for } x \in \hat{N}_k, \\ 1, & \text{for } x \in M_\sigma \setminus \bigcup_{k=1}^{p} \hat{N}_k, \end{cases} \tag{4.14}$$

and $\|\cdot\|_{m,\alpha}$ is the standard Hölder norm introduced in Section 3.2 and used in the previous chapter.

Note that $W(x) > 1$ for all $x \in M_\sigma$, and therefore $W^{-1}(x) < 1$. Moreover, $W(x)$ attains its minimum at $r = \sqrt{R_k}$, which lies roughly at the centre of the region R_k (this relates to the points c_k mentioned earlier). The plot below illustrates the behaviour of $W(x)$ for $x \in N_k$ when $R_k = 4$.



Remark 4.3.1. Note that $W(x)$ is not continuous at the torus T_k, where the gluing takes place as, at $r = R_k$, there is a jump between 1 and $1 + R_k^{-0.1}$. However, this discontinuity is not essential as $R_k > 2^{1/(n-1)}$ by the gluing construction described in Section 3.1.2.

This norm successfully *isolates* the Einstein trivial variations, as demonstrated by the following result.

Lemma 4.3.1. *Let h be a symmetric bilinear form that is T^{n-1} invariant. Then*

$$|h| < r^{0.1} + r^{-0.1} \quad \text{and} \quad L_{g_{hyp}}(h) = 0$$

implies that $h = 0$.

Proof. The result follows directly from the set of fundamental solutions to the system of ODEs 3.12 and the bound on h. □

Einstein trivial variations are the *only* fundamental solutions to the system of ODEs 3.12 (see equation 4.7) that *can not* be bounded above by $r^{0.1}+r^{-0.1}$ when working with weighted Hölder norms. Thus, by working with $\bar{h}$, the component of h orthogonal to the Einstein trivial variations, we are able to use this upper bound to conclude that $\bar{h} = 0$. As a result, we obtain a bound on L^{-1} that is independent of $R_{\max}$, on this complement. We then just need to define a hybrid norm $\|\cdot\|_{**}$ that uses weighted Hölder norms in $\bar{h}$ and a different norm for u_k:

$$\|h\|_{m,\alpha;**} := \inf_{\bar{h},\, u_1,\ldots,u_p} \left(\|\bar{h}\|_{m,\alpha;*} + \sum_{k=1}^{p} |u_k| \right).$$

4.3.3 Proof of the uniform bound

We are now ready to prove the main Theorem of this section:

Theorem 5. *For M_σ be the manifold constructed by Dehn Filling, as described in Chapter 3. For σ sufficiently large, there is a constant Λ such that the operator L invertible and*

$$\|h\|_{m,\alpha;**} \leq \Lambda \|L_{\bar{g}} h\|_{m-2,\alpha;*} \tag{4.15}$$

for any symmetric bilinear form h on M_σ.

Proof. The proof follows the same strategy as in Theorem 4. Namely, we assume that

$$\|h^i\|_{0;**} = 1, \quad \text{and} \quad \|L^i(h^i)\|_{m-2,\alpha;*} \to 0, \tag{4.16}$$

and aim to reach a contradiction by analysing the behaviour of h^i in the three distinct regions of the manifold: the core, the cusp, and the black-hole region. The fact that

$\|\bar{h}^i\|_{0;*} \to 0$ as $\|L^i(h^i)\|_{m-2,\alpha;*} \to 0$ follows almost directly from Lemma 4.3.1, after using the weighted Hölder norm to upper bound $|\bar{h}| < r^{0.1} + r^{-0.1}$.

We will thus focus on the behaviour of $|u_k^i|$ as $\|L^i(h^i)\|_{m-2,\alpha;*} \to 0$. For convenience, we will now denote $h = u_k$. We aim to show that, under the assumptions of this theorem,

$$\|h\|_{m,\alpha} \le \Lambda \|L_{\bar{g}} h\|_{m-2,\alpha;*}.$$

Notice the similarity between this inequality and the one in Theorem 4. In this case, the $\log R_{\max}$ factor is no longer necessary, but this comes at the cost of working with the stronger weighted norm. Again, assuming the claim is false, we consider a sequence

$$\|h^i\|_{L^\infty} = 1, \quad \text{and} \quad \|L^i(h^i)\|_{C^{m-2,\alpha;*}} \to 0, \tag{4.17}$$

and analyse it across the three regions of the manifold.

The key difference from the proof of Theorem 4 lies in the analysis of Case 2 (the Cusp Case), and in particular, in Lemma 4.2.1. Therefore, rather than repeating the entire proof, we will focus on re-establishing this result using the new bound on $|f_{ab}^i(r)|$, as formalized in Lemma 4.3.2.

Lemma 4.3.2. *Let $h \in S_2^{m,\alpha}$ on $\mathbb{R}^2 \times \mathbb{R}^{n-2}$ with the black-hole metric and satisfying*

$$\|L^i(h^i)\|_{C^{m-2,\alpha,*}} \to 0$$

defined in the sequence of Manifolds (M^i, g^i, x^i) such that $dist_{\bar{g}}(x^i, (T_k^{n-2})^i) \to \infty$ and $dist_{\bar{g}}(x^i, y_0) \to \infty$ (i.e Cusp case). Then $h^i \to 0$.

Proof. We proceed, as before, by considering the averaged values of $h_{ab}^i(r)$ and $f_{ab}^i(r)$ over the torus, and the corresponding ODE:

$$r^2(h_{ab}^i(r))'' + nr(h_{ab}^i(r))' = f_{ab}^i(r) + e_{ab}^i(r), \tag{4.18}$$

where $e_{ab}^i(r)$ is the error term arising from averaging and the difference between the metrics

$$e_{ab}^i(r) = O\left(\frac{r}{R_k^i}\right) + O\left(r^{-(n-1)}\right).$$

The difference in this argument is that we now have an explicit upper bound for $|f_{ab}^i(r)|$. Setting $\alpha^i := \|f_{ab}^i\|_{m-2,\alpha;*}$, we obtain:

$$|f_{ab}^i(r)| < \alpha^i \left(\left(\frac{r}{R_k^i}\right)^{0.1} + r^{-0.1}\right),$$

and thus,

$$|f^i_{ab}(r)| = O\left(\alpha^i\left(\frac{r}{R^i_k}\right)^{0.1}\right) + O\left(\alpha^i r^{-0.1}\right).$$

From expression (4.10) in Lemma 4.2.1, we know that

$$|h'^i_{ab}(r)| \leq \frac{1}{r^n}\int_{C_0}^{r} \left|t^{n-2}\left(f^i_{ab}(t) + e^i_{ab}(t)\right)\right| dt.$$

The integral involving the error term (term B in 4.10) proceeds exactly as in the earlier case. However, integrating the first term (term A in 4.10) with the new bound on f^i_{ab}, we have

$$\frac{1}{r^n}\int_{C_0}^{r}\left|t^{n-2}f^i_{ab}(t)\right| dt \leq \frac{1}{r^n}\int_{C_0}^{r}\left(\alpha^i\frac{t^{n-1.9}}{R^i_k} + \alpha^i t^{n-2.1}\right) dt < \alpha^i\left(\frac{r^{-0.9}}{R^i_k}\right) + \alpha^i r^{-1.1}.$$

Combining this with the bound from the integration of the error term (see 4.11), we obtain for sufficiently large r,

$$|h'^i_{ab}(r)| \leq C\left(\alpha^i\left(\frac{r^{-0.9}}{R^i_k}\right) + \alpha^i r^{-1.1} + \frac{1}{R^i_k} + \frac{\log r}{r^n}\right).$$

Integrating once more from r to r^i, we find:

$$|h^i_{ab}(r) - h^i_{ab}(r^i)| \leq C\left(\alpha^i\left(\frac{(r^i)^{0.1}}{R^i_k} + (r^i)^{-0.1}\right) + \frac{r^i}{R^i_k} + (r^i)^{-(n-1)}\log r^i + r^{-(n-1)}\log r\right),$$

which leads to

$$|h^i_{ab}(r)| \leq C\alpha^i + Cr^{-(n-1)}\log r + |h^i_{ab}(r^i)| \leq Cr^{-(n-1)}\log r, \tag{4.19}$$

uniformly on the interval $[C_0, r^i]$, since $\alpha^i \to 0$ and $|h^i_{ab}(r^i)| \to 0$ as $i \to \infty$, by Case 1 of Lemma 4.2.1.

Therefore, as $r \to \infty$, we have $|h^i_{ab}(r)| \to 0$, completing the proof of Lemma 4.3.2.

□

Now, setting $h = u_k$, we obtain a contradiction to the assumption in (4.16), thus completing the proof of the theorem.

□

Remark 4.3.2. Instead of defining W with the specific value 0.1 as in (4.14), this proof can be generalized to an arbitrary choice of W, as long as the decay rate of $\frac{1}{W}$ is smaller than the absolute value of the non-zero growth rates of the fundamental solutions of the system of ODEs 3.12, as already done by [Biq07; HJ22]..

4.3.4 Proof of Theorem 1

Fortunately, we have eliminated the dependence on $\log R_{\max}$ in the bound 4.4. However, the trade-off is that the norms on both sides of the inequality are now different. In order to apply Theorem 3, we must show that the Einstein trivial deformations, which make the difference between the two norms, have only a weak influence on the nonlinear terms.

Once this has been established, we need to reprove the Condition 4.1 using the weighted Hölder norm 4.5 (condition 4.3 follows directly) to be able to apply the Banach Fixed Point Theorem. These technical details are omitted.

Chapter 5

Numerical approximation of the Einstein metric

5.1 Physics-informed neural networks

In Chapter 4, we proved the existence of an Einstein metric, starting from the approximation in Chapter 3. We now ask:

Can we numerically approximate this Einstein metric?

We approach this question using physics-informed neural networks (PINNs). These have recently succeeded in constructing Calabi–Yau and G_2 metrics, with a lot of research carried out in this context: [AYHB20; Lar+22; Mag+21; GK23; Vis+22; MSY20; DPQ24]. However, it is only very recently that these methods have been applied to negatively curved Einstein manifolds ([HGS24; Luc25]).

The broad idea behind PINNs is to represent the unknown metric g that solves the Einstein equation $\mathrm{Ric} = \lambda g$ as a neural network $g \sim \mathcal{N}(x; \theta)$, where x are points sampled from the manifold. The network is then trained by minimizing a loss function that incorporates both the Einstein equation, boundary and regularity conditions. In our setting, these will only be smoothness of the metric. The loss function thus takes this schematic form:

$$L(\theta) = \text{Einstein equation residuals} + \text{Regularity terms}. \tag{5.1}$$

There are multiple optimization algorithms that one can use to train this neural network, among the most popular ones are stochastic gradient descent, ADAM ([KB15]), or momentum-based methods ([Qia99]).

In this essay, we follow closely the work of ([Luc25]), as a neural network is used to approximate the Einstein metric following the same procedure explained in Chapter 3 and Chapter 4. That is, first the network is *pre-trained* to approximate the nearly Einstein metric from Chapter 3. As we do have a close form of this metric (Equation 3.6), this step only involves supervised learning. The next step is to *learn* the perturbation of this metric into an Einstein one, as described in Chapter 4. This time it is unsupervised learning as we don't have a close form solution. The pre-training step ensures that the optimizations starts *close* to the exact solution, which is crucial to ensure the optimization process converges. [Luc25] then trains the neural network on a 3-dimensional manifold where one of its cusps has been filled with a solid torus endowed with the black-hole metric. This manifold has been constructed explicitly using Coxeter polytopes, with the approach described in ([ERT12; PV05]). The important point is that we have a *global coordinate system* to sample points on the manifold and define the Riemannian metric in each region. This architecture achieves state-of-the art results on the Einstein metric found. For more details on the architecture and results refer to [Luc25].

5.2 Loss landscape and stability of Einstein metrics

In this dissertation, we build upon the work of [Luc25] and analyse how properties of the moduli space of Einstein metrics are reflected in the loss surface $L(\theta)$ of this PINN. In particular, we will focus on the *stability* of Einstein metrics. This is usually studied via the Einstein–Hilbert functional $S : M \to \mathbb{R}$:

$$S(g) = \int_M \mathrm{scal}_g \, dV_g,$$

where scal_g is the scalar curvature of the metric g, and dV_g is the corresponding volume form. It is well-known that Einstein metrics are saddle points of S. This motivates the following notion of stability [Krö15]:

Definition 8. *An Einstein metric g is stable if the second variation $S''(h) \leq 0$ for all h satisfying $\mathrm{tr}_g\, h = 0$ and $\mathrm{div}_g\, h = 0$. Such tensors are called transverse-traceless (TT).*

To *numerically* explore the stability of Einstein metrics, we analyse the eigenvalue spectrum of the Hessian of the loss surface of the trained PINN from [Luc25]. The eigenvalues provide a notion of curvature, where a flatter spectrum indicates higher stability against perturbations. We begin by visualizing the loss landscape along two randomly chosen directions. As shown in Figure 5.1, the approximate Einstein metric seems to lie in a sharp minimum.

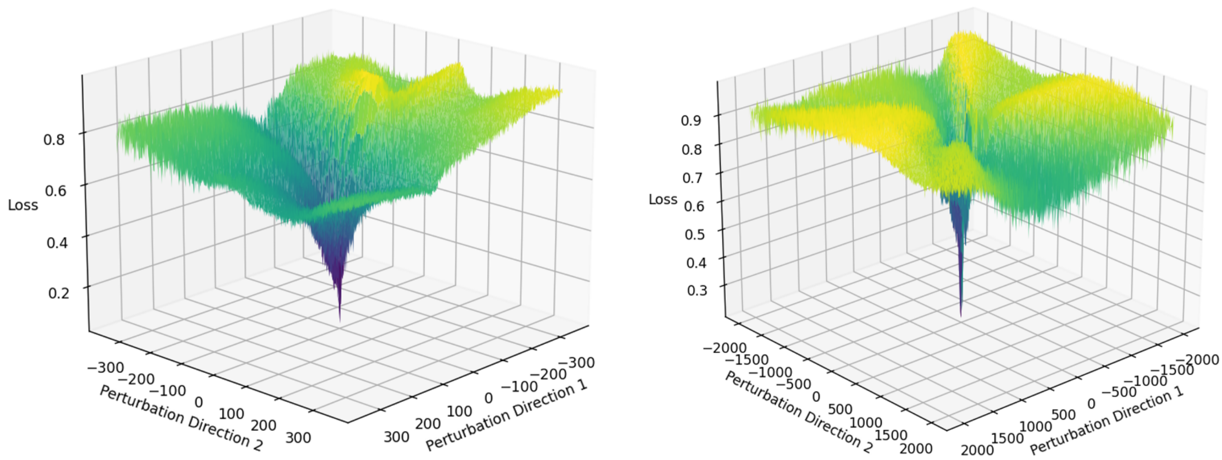


Figure 5.1: Loss surface along two random directions

While this is just a projection of a very high dimensional surface, and thus insufficient to reach any conclusion, the wide range of eigenvalues shown in Figure 5.2 confirms this instability.

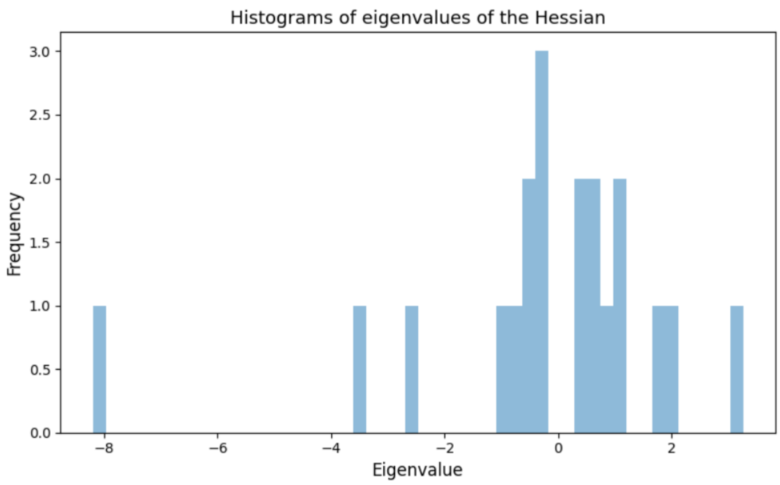


Figure 5.2: 50 larger and 50 smaller eigenvalues of the Hessian of $L(\theta)$

This points out a possible challenge in the training dynamics, as sharper minimum are harder to detect and require more sophisticated optimization algorithms such as sharpness-aware minimization [For+21].

From the previous theoretical discussion, we hypothesize that restricting the set of perturbations that the neural network explores to those which are transverse-traceless, may lead to a flatter minimum. To test this idea, we introduce a *TT regularizer* into the loss function, which softly constraints the network's search space by only considering transverse-traceless perturbations. The modified loss function then takes the schematic form

$$L(\theta) = \text{Einstein residuals} + \text{Regularity terms} + \text{TT regularization}, \tag{5.2}$$

where the TT regularization term penalizes both the trace and divergence:

$$\mathrm{TT}(h) := \lambda_1 \| \operatorname{tr}_{\bar{g}} h\|^2 + \lambda_2 \| \operatorname{div}_{\bar{g}} h\|^2,$$

given the approximated metric $\bar{g}$ obtained in the pre-training phase and the hyperparameter λ_1 and λ_2. The full Python implementation of this regularizer can be found in https://anonymous.4open.science/r/TT-regularizer–2B57/README.md.

We see in Figure 5.3 that this regularization indeed reduces the complexity of the loss surface by promoting a flatter spectrum, achieving the desired effect.

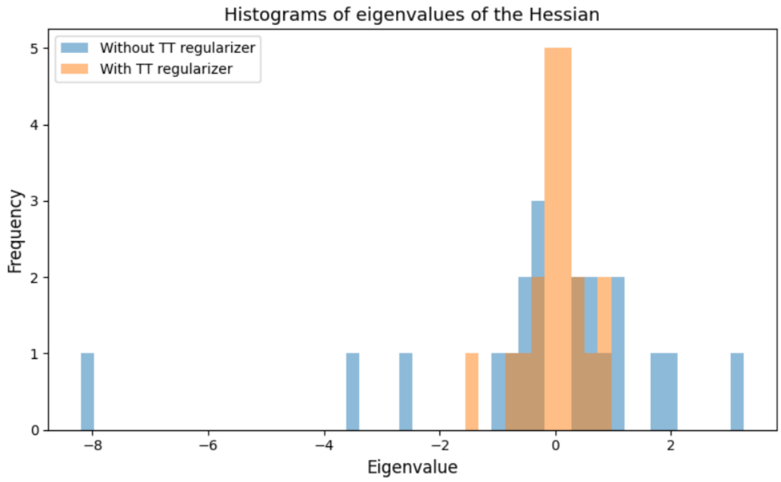


Figure 5.3: 50 larger and 50 smaller eigenvalues of the Hessian before and after implementing the TT regularizer

Building up from these ideas, it would be interesting to construct architectures that are invariant under the choice of local coordinates. These types of architectures have already been successfully implemented in other areas of Geometric Deep Learning (see [Wei+25]). This could uncover insights into the moduli space of Einstein metrics, especially in higher dimensions. For instance, it could allow us to study rigidity properties of Einstein metrics. In this example, since hyperbolic metrics are rigid, we would expect the loss surface to exhibit a unique minimum under such a setup. The architecture introduced by [Luc25] is also very limited in terms of generalization, as it relies on explicit representations of the input manifold and the existence of a global coordinate chart. The work of [HGS24] overcomes this limitation by defining neural architectures over overlapping coordinate charts instead.

Chapter 6

Conclusion

In this essay, we analysed the construction of negative Einstein metrics by a generalized version of Dehn Filling. We began by following the first proof of Theorem 1 by [And06]. Along the way, we encountered a particular type of bounded deformations of the cusp metric that complicated the analysis and, in Section 4.3, we analyse an argument by [Bam12] where a hybrid norm is used to handle these variations differently, resolving the technical challenges faced in [And06]. More recently, the proof has been refined by [HJ22], who established the result for manifolds with arbitrarily large volume and only requiring exponential control on the metric along the cusp. This last assumption was further relaxed by [Jä23] by allowing a weaker exponential control. [Jä23] used an axiomatic framework to prove this result, which I think it would be interesting to study in future work as a possible generalization of this dissertation. I would also be interested in exploring other gluing constructions for obtaining approximate metrics, beyond the Dehn filling studied in this essay.

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